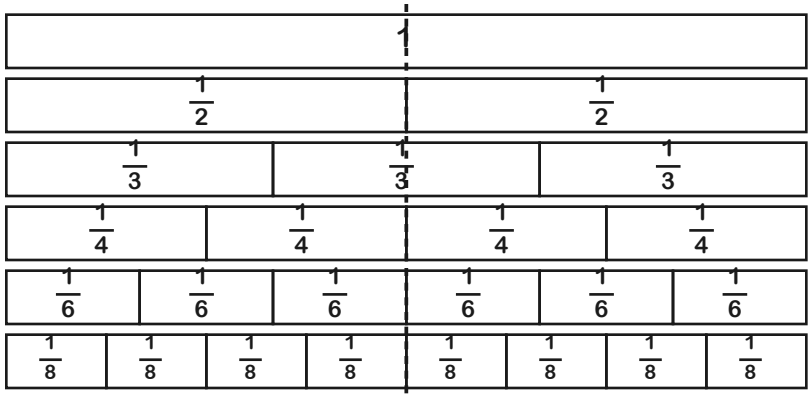
Equal Fractions

Different pieces, same amount.

The fraction wall

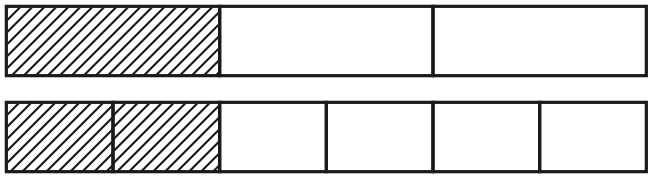


Follow the dashed line:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$

Same length, so they are equal fractions.

Cut every piece in two

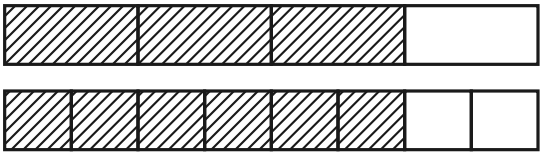


$$\frac{1}{3} \quad \frac{2}{6}$$

Twice as many pieces, each half as big.

$$\frac{1}{3} = \frac{2}{6}$$

Your turn!



$$\frac{3}{4} = \frac{\boxed{6}}{8}$$

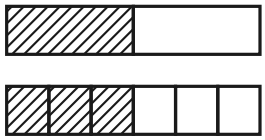
For teachers

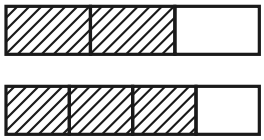
Goal: equal fractions name the same amount of the same whole (3.NF.3a–b). Line the strips up: "Is the shaded part the same length?" Cutting every piece in two doubles the top and the bottom.

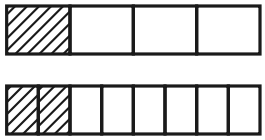
Equal Fractions: Practice

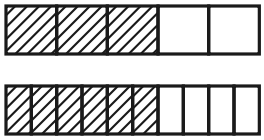
Same whole, same amount shaded: the fractions are equal.

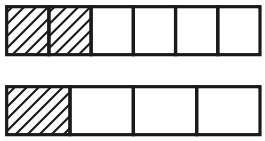
Part 1 Same amount? Same whole. Is the same amount shaded? Circle yes or no.

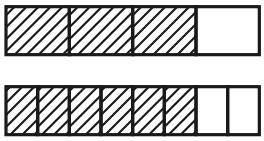
1) $\frac{1}{2}$ $\frac{3}{6}$  (yes) no

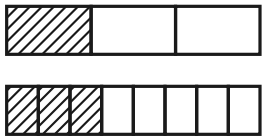
2) $\frac{2}{3}$ $\frac{3}{4}$  yes (no)

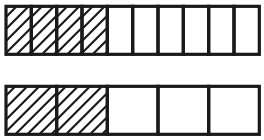
3) $\frac{1}{4}$ $\frac{2}{8}$  (yes) no

4) $\frac{3}{5}$ $\frac{6}{10}$  (yes) no

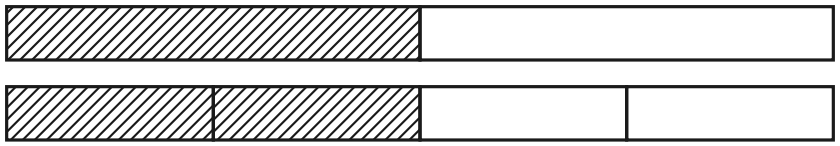
5) $\frac{2}{6}$ $\frac{1}{4}$  yes (no)

6) $\frac{3}{4}$ $\frac{6}{8}$  (yes) no

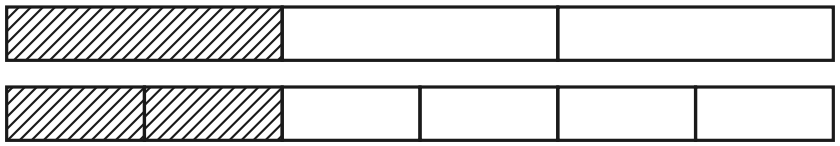
7) $\frac{1}{3}$ $\frac{3}{8}$  yes (no)

8) $\frac{4}{10}$ $\frac{2}{5}$  (yes) no

Part 2 Line them up Count the shaded pieces in the second strip.

9) 

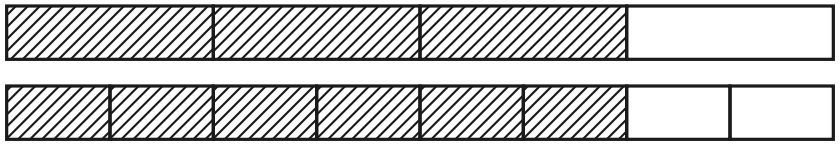
$\frac{1}{2} = \frac{\boxed{2}}{4}$

10) 

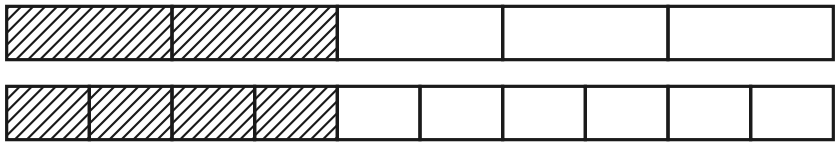
$\frac{1}{3} = \frac{\boxed{2}}{6}$

Equal Fractions: Practice (continued)

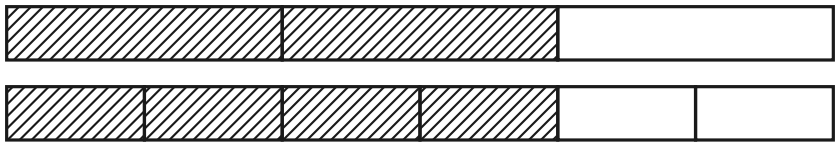
Part 2 Line them up (continued)

11) 

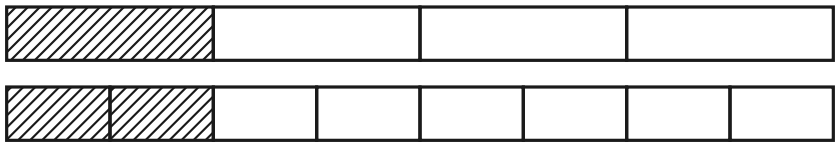
$\frac{3}{4} = \frac{6}{8}$

12) 

$\frac{2}{5} = \frac{4}{10}$

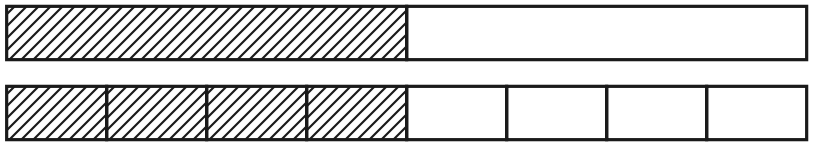
13) 

$\frac{2}{3} = \frac{4}{6}$

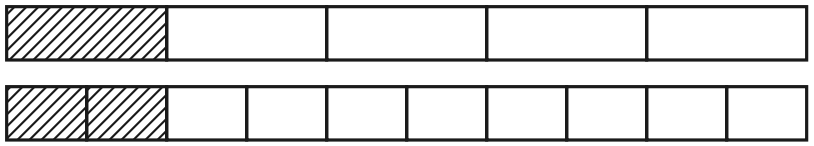
14) 

$\frac{1}{4} = \frac{2}{8}$

Part 3 Shade to match Shade the same amount in the second strip, then write the fraction.

15) 

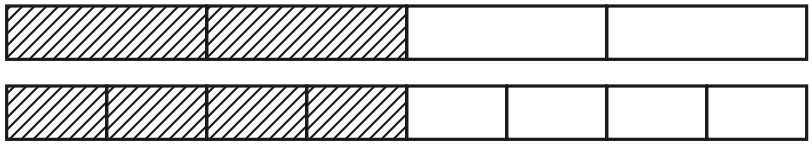
$\frac{1}{2} = \frac{4}{8}$

16) 

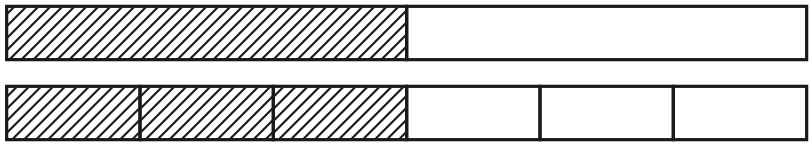
$\frac{1}{5} = \frac{2}{10}$

Equal Fractions: Practice (continued)

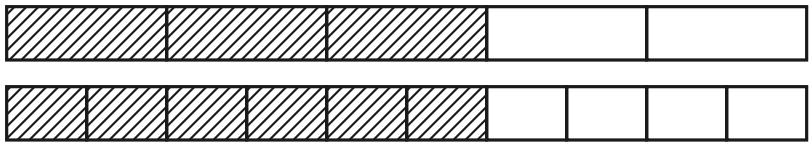
Part 3 Shade to match (continued)

17) 

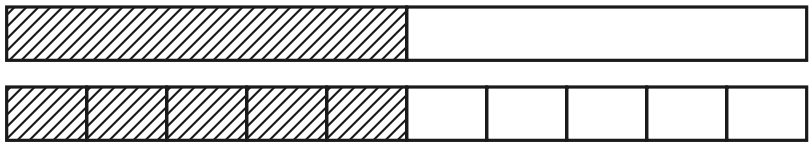
$\frac{2}{4} = \frac{\boxed{4}}{8}$

18) 

$\frac{1}{2} = \frac{\boxed{3}}{6}$

19) 

$\frac{3}{5} = \frac{\boxed{6}}{10}$

20) 

$\frac{1}{2} = \frac{\boxed{5}}{10}$

Part 4 Find the top number No help this time. Picture the wall.

21) $\frac{1}{2} = \frac{\boxed{4}}{8}$

22) $\frac{2}{3} = \frac{\boxed{4}}{6}$

23) $\frac{1}{4} = \frac{\boxed{2}}{8}$

24) $\frac{4}{5} = \frac{\boxed{8}}{10}$

25) $\frac{1}{2} = \frac{\boxed{5}}{10}$

26) $\frac{3}{4} = \frac{\boxed{6}}{8}$

Equal Fractions: Practice (continued)

Part 4 Find the top number (continued)

27) $\frac{1}{5} = \frac{\boxed{2}}{10}$

28) $\frac{1}{3} = \frac{\boxed{2}}{6}$

29) $\frac{2}{4} = \frac{\boxed{4}}{8}$

30) $\frac{3}{5} = \frac{\boxed{6}}{10}$

31) $\frac{1}{2} = \frac{\boxed{3}}{6}$

32) $\frac{2}{5} = \frac{\boxed{4}}{10}$

Part 5 Fewer, bigger pieces No help this time. Same amount with bigger pieces.

33) $\frac{4}{8} = \frac{\boxed{1}}{2}$

34) $\frac{2}{6} = \frac{\boxed{1}}{3}$

35) $\frac{6}{8} = \frac{\boxed{3}}{4}$

36) $\frac{8}{10} = \frac{\boxed{4}}{5}$

37) $\frac{2}{8} = \frac{\boxed{1}}{4}$

38) $\frac{5}{10} = \frac{\boxed{1}}{2}$

Equal Fractions: Practice (continued)

Part 5 Fewer, bigger pieces (continued)

39) $\frac{4}{6} = \frac{\boxed{2}}{3}$

40) $\frac{6}{10} = \frac{\boxed{3}}{5}$

41) $\frac{3}{6} = \frac{\boxed{1}}{2}$

42) $\frac{2}{10} = \frac{\boxed{1}}{5}$

43) $\frac{2}{4} = \frac{\boxed{1}}{2}$

44) $\frac{4}{10} = \frac{\boxed{2}}{5}$

Part 6 Find the bottom number No help this time.

45) $\frac{1}{2} = \frac{2}{\boxed{4}}$

46) $\frac{1}{3} = \frac{2}{\boxed{6}}$

47) $\frac{3}{4} = \frac{6}{\boxed{8}}$

48) $\frac{1}{5} = \frac{2}{\boxed{10}}$

49) $\frac{2}{3} = \frac{4}{\boxed{6}}$

50) $\frac{1}{2} = \frac{4}{\boxed{8}}$

Equal Fractions: Practice (continued)

Part 6 Find the bottom number (continued)

51) $\frac{4}{8} = \frac{1}{\boxed{2}}$

52) $\frac{6}{10} = \frac{3}{\boxed{5}}$

53) $\frac{2}{8} = \frac{1}{\boxed{4}}$

Part 7 Halves in disguise No help this time. Each one is the same as 1/2.

54) $\frac{2}{4} = \frac{\boxed{3}}{6}$

55) $\frac{3}{6} = \frac{\boxed{4}}{8}$

56) $\frac{4}{8} = \frac{\boxed{5}}{10}$

57) $\frac{2}{4} = \frac{\boxed{5}}{10}$

58) $\frac{3}{6} = \frac{\boxed{5}}{10}$

59) $\frac{5}{10} = \frac{\boxed{4}}{8}$

Part 8 Keep going No help this time.

60) $\frac{1}{2} = \frac{\boxed{2}}{4} = \frac{\boxed{4}}{8}$

61) $\frac{3}{6} = \frac{\boxed{2}}{4} = \frac{\boxed{1}}{2}$

Equal Fractions: Practice (continued)

Part 8 Keep going (continued)

62) $\frac{1}{2} = \frac{\boxed{3}}{6} = \frac{\boxed{5}}{10}$

63) $\frac{8}{8} = \frac{\boxed{4}}{4} = \frac{\boxed{2}}{2}$

64) $\frac{1}{2} = \frac{\boxed{2}}{4} = \frac{\boxed{3}}{6} = \frac{\boxed{4}}{8}$

65) $\frac{2}{2} = \frac{\boxed{3}}{3} = \frac{\boxed{5}}{5}$

Part 9 Make it simpler

66) $\frac{3}{9} = \frac{\boxed{1}}{3}$

67) $\frac{6}{9} = \frac{\boxed{2}}{3}$

68) $\frac{3}{12} = \frac{\boxed{1}}{4}$

Part 10 Equal fractions

69) $\frac{1}{3} = \frac{\boxed{3}}{9}$

70) $\frac{2}{3} = \frac{\boxed{6}}{9}$

71) $\frac{2}{6} = \frac{\boxed{4}}{12}$

Equal Fractions: Practice (continued)

Part 11 Remember? Problems from earlier packets.

72) Round 655 to the nearest ten: 660

73) 8 m by 6 m: area = 48 sq m

74) three fifths = $\frac{3}{5}$

75) $\frac{10}{2}$ = 5

76) $\frac{7}{4}$ = 1 whole and $\frac{3}{4}$

77) 8 groups of 6 = 48

Part 12 Word problems Draw strips to check.

78) Ana ate 2/4 of a pizza. Ben ate 1/2 of a same-size pizza. Did they eat the same amount?

Number sentence 2/4 = 1/2

Answer yes

Draw or show your work

79) A ribbon has 8 equal parts. Mia uses 4 of them. What fraction of the ribbon is that, in halves?

Number sentence 4/8 = 1/2

Answer 1/2 of the ribbon

Draw or show your work

80) Leo shades 1/3 of a bar. Kim's same-size bar has 6 parts. How many should Kim shade?

Number sentence 1/3 = 2/6

Answer 2 parts

Draw or show your work

Equal Fractions: Practice (continued)

Part 12 Word problems (continued)

81) A cake is cut into 10 equal slices. Half of the cake is eaten. How many slices is that?

Number sentence 1/2 = 5/10

Answer 5 slices

Draw or show your work

82) Challenge: Sam ran 3/4 of a mile. Jo ran 6/8 of a mile. Who ran farther?

Number sentence 3/4 = 6/8

Answer same distance

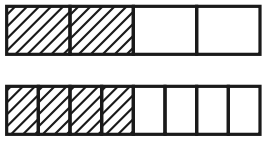
Draw or show your work

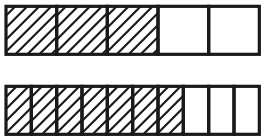
Check-In: Equal Fractions

Show what you know. No help this time.

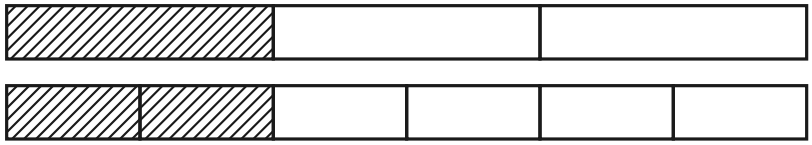
Mastery: 8 of 9 (80%)

Part 1 Same amount? Circle yes or no.

1) $\frac{2}{4}$ $\frac{4}{8}$  (yes) no

2) $\frac{3}{5}$ $\frac{7}{10}$  yes (no)

Part 2 Shade to match

3)  $\frac{1}{3} = \frac{2}{6}$

Part 3 Find the top number

4) $\frac{4}{8} = \frac{2}{4}$

5) $\frac{1}{2} = \frac{2}{4}$

6) $\frac{2}{10} = \frac{1}{5}$

Part 4 Find the bottom number

7) $\frac{4}{6} = \frac{2}{3}$

8) $\frac{2}{4} = \frac{1}{2}$

9) $\frac{1}{2} = \frac{5}{10}$